

MAD 2

Trekking Management System

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I am pursuing the BS Degree in Data Science and Applications at IIT Madras. This project was developed as part of the Modern Application Development II course to demonstrate the design and implementation of a complete full-stack web application using Flask, Vue.js, REST APIs, SQLAlchemy, Redis and Celery.

Description

The Trekking Management System is a role-based web application designed to simplify the management of trekking activities. The system allows administrators to manage treks, staff and users, staff members to manage their assigned treks and participants, and users to browse, search and book trekking packages while maintaining their booking history.

Technologies Used

Frontend

- Vue.js 3
- Vue Router
- Pinia
- Axios
- Bootstrap 5
- Bootstrap Icons

The frontend follows a component-based architecture with reusable layouts, tables, modals and search components to provide a responsive and user-friendly interface.

Backend

- Flask
- Flask-Login
- Flask-SQLAlchemy
- Flask-Marshmallow
- Flask-Migrate

Flask is used to implement REST APIs, authentication, authorization and business logic. SQLAlchemy ORM simplifies database interactions while Marshmallow handles object serialization.

Database

- SQLite

SQLite stores all application data including users, treks and bookings.

Background Services

- Redis
- Celery

Redis is used both as the caching layer and as the Celery broker/result backend. Celery executes asynchronous tasks such as booking history export, reminder emails and scheduled reports.

Database Schema Design

The application uses three primary entities.

User

Stores all registered users, staff members and administrators.

Main fields include:

- id
- full_name
- email
- password
- contact_number
- role
- specialization
- experience
- status

The role attribute differentiates Admin, Staff and User accounts while the status field supports activation and blacklisting.

Trek

Stores trekking package information.

Main fields include:

- id
- trek_name
- description
- location
- difficulty
- duration
- available_slots
- booked_slots
- start_date
- end_date
- status
- assigned_staff_id

Each trek is assigned to one staff member while multiple users can book the same trek.

Booking

Maintains booking records between users and treks.

Main fields include:

- id
- user_id
- trek_id
- booking_date
- booking_status
- remarks

A composite uniqueness constraint prevents duplicate bookings for the same user and trek combination while preserving booking history.

API Design

The backend exposes RESTful APIs grouped according to user roles.

Authentication APIs

- User Registration
- Login
- Logout

Admin APIs

- Dashboard
- Trek CRUD operations
- Staff Management
- User Management
- Assign Staff
- Search functionality

Staff APIs

- Dashboard
- Assigned Treks
- Update Trek
- Complete Trek
- View Participants

User APIs

- Browse Available Treks
- Search Treks
- Book Trek
- Cancel Booking
- Booking History
- Export Booking History
- Profile Management

The APIs exchange JSON responses and are protected using role-based decorators to ensure only authorized users can access specific resources.

Architecture and Features

The application follows a layered architecture.

The backend is organized into controllers, routes, models, schemas, decorators, tasks and utility modules. Controllers contain business logic while routes define API endpoints. SQLAlchemy models manage database entities and Marshmallow schemas serialize responses. Celery tasks handle long-running background operations, and Redis provides caching for frequently accessed data.

The frontend follows a component-based structure using Vue.js. Reusable components such as DashboardLayout, SearchBar, Loader, EmptyState, DataTable and modal dialogs reduce duplication and maintain UI consistency across Admin, Staff and User modules.

The following major features have been implemented:

Authentication

- Role-based login
- Protected routes
- Session management
- User and staff blacklisting

Admin Module

- Dashboard summary
- Trek management
- Staff management
- User management
- Search functionality
- Trek assignment

Staff Module

- Dashboard
- Assigned trek management
- Slot management
- Trek status updates
- Trek completion
- Participant management

User Module

- Browse available treks
- Search and filter treks by difficulty, location and duration
- Trek booking
- Duplicate booking prevention
- Booking history
- Booking cancellation

- CSV export of booking history
- Profile management

Additional Features

- Redis caching for dashboards and available treks
- Celery-based asynchronous booking export
- Daily reminder and monthly report task support
- Toast notifications
- Loading indicators
- Empty state components
- Responsive Bootstrap interface

Video

Video Demonstration Link:

https://drive.google.com/file/d/1Ke7ODbJL8yV_kYFIS-jVcf-fytLJlde3/view?usp=drive_link